

- **Prompt as a deployable artifact:** When a prompt changes, behaviour changes — but unlike code, there is no compile-time error. Observability must correlate output quality with the prompt version.

Observability for AI is not an extension of APM (application performance monitoring) — it is a new discipline that requires new instrumentation primitives, new quality signals, and new feedback loops between production telemetry and model evaluation pipelines.

The role of different open source tools in observability

Tool / Project	Category	Role in AI observability stack
OpenTelemetry (OTel)	Observability	Universal instrumentation standard for traces, metrics, logs — backbone of any AI observability stack
Prometheus	Metrics	Time-series metrics store — captures LLM latency, token throughput, error rates, GPU utilisation
Grafana	Visualisation	Unified dashboards for metrics, logs, and traces — industry standard for observability UI and alerting
Grafana Loki	Logging	Log aggregation optimised for Kubernetes-native environments — low-cost, label-indexed log querying
Jaeger	Distributed tracing	End-to-end trace visualisation for LLM chains and agentic workflows — OpenTelemetry compatible
Tempo (Grafana)	Tracing backend	High-scale, cost-efficient trace storage integrated natively with Grafana and Loki
LangFuse	LLM observability	Purpose-built for LLM: prompt versioning, token cost tracking, eval scoring, session replay
OpenLLMetry	LLM instrumentation	OpenTelemetry-based SDK — auto-instruments LangChain, OpenAI, Anthropic, Hugging Face
Helicone (OSS)	LLM proxy + Obs	Reverse proxy for LLM APIs capturing latency, cost, errors with zero code change
Traceloop	LLM pipelines	Workflow-level observability for multi-step LLM chains; integrates with OTel and existing stacks
MLflow	ML/LLM lifecycle	Experiment tracking, model registry, and LLM run logging — strong for fine-tuning and eval pipelines
Evidently AI	Model monitoring	Data drift, model quality, and LLM output quality monitoring — real-time and batch evaluation
Arize Phoenix	LLM evaluation	Open source LLM observability — trace visualisation, retrieval evaluation, embedding drift detection
Vector Admin (OSS)	Vector DB obs	Observability layer for vector databases (Weaviate, Qdrant, Chroma, Pinecone)
Weaviate / Qdrant	Vector storage	Open source vector DBs — monitorable via Prometheus exporters for retrieval latency and index health
Alertmanager	Alerting	Prometheus-native alerting with routing, deduplication, and silencing — essential for SLA enforcement
OpenCost	Cost observability	Real-time Kubernetes cost monitoring — extended to track LLM inference cost per workload
LangSmith (OSS)	LLM tracing	LangChain-native chain-level trace capture and debugging; community version available

The open source imperative for AI observability

Vendor-native observability for AI is maturing — but it comes with lock-in, opaque pricing, and limited extensibility. When your LLM observability stack is built into your model provider's console, you have observability only for that provider. Open source changes the equation on three axes:

- **Vendor neutrality:** Instrument Anthropic, OpenAI, Mistral, locally-hosted Ollama models, and Hugging Face endpoints with the same stack — OpenLLMetry and OpenTelemetry abstract the differences.